Intensity-Dependent tACS Entrainment Effects in a Cortical Microcircuit: A Computational Study

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Introduction

Transcranial alternating current stimulation (tACS) and its limitations

- **Neural entrainment:** Synchronization of neuronal spiking activity shifting towards a specific phase of the tACS waveform
- The efficacy of tACS-induced entrainment remains debated due to inconsistent outcomes

▶ Previous studies investigating the cellular mechanism of tACS

- Experimental and computational studies revealed that **neural morphology** is a crucial factor in neural responses by tACS [1]
- Some studies addressed the importance of implementing **synaptic connections** within the computational model [2]

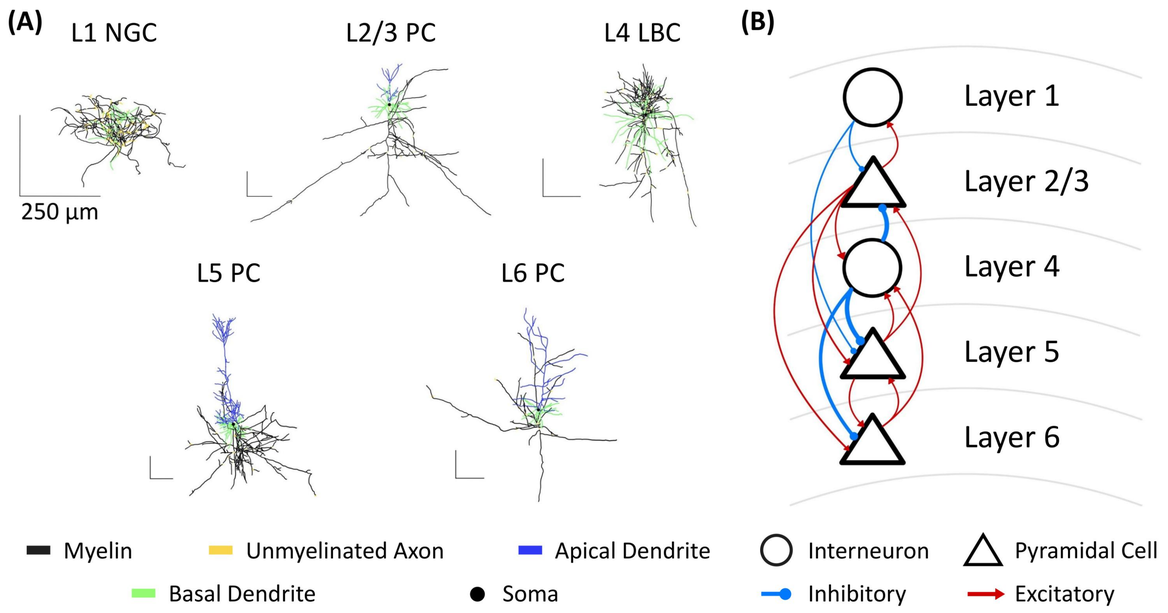
Objectives

- Investigate tACS-induced neural entrainment using a synaptically connected cortical microcircuit model
- Analyze how individual neuron models influence neural oscillations and entrainment in response to tACS

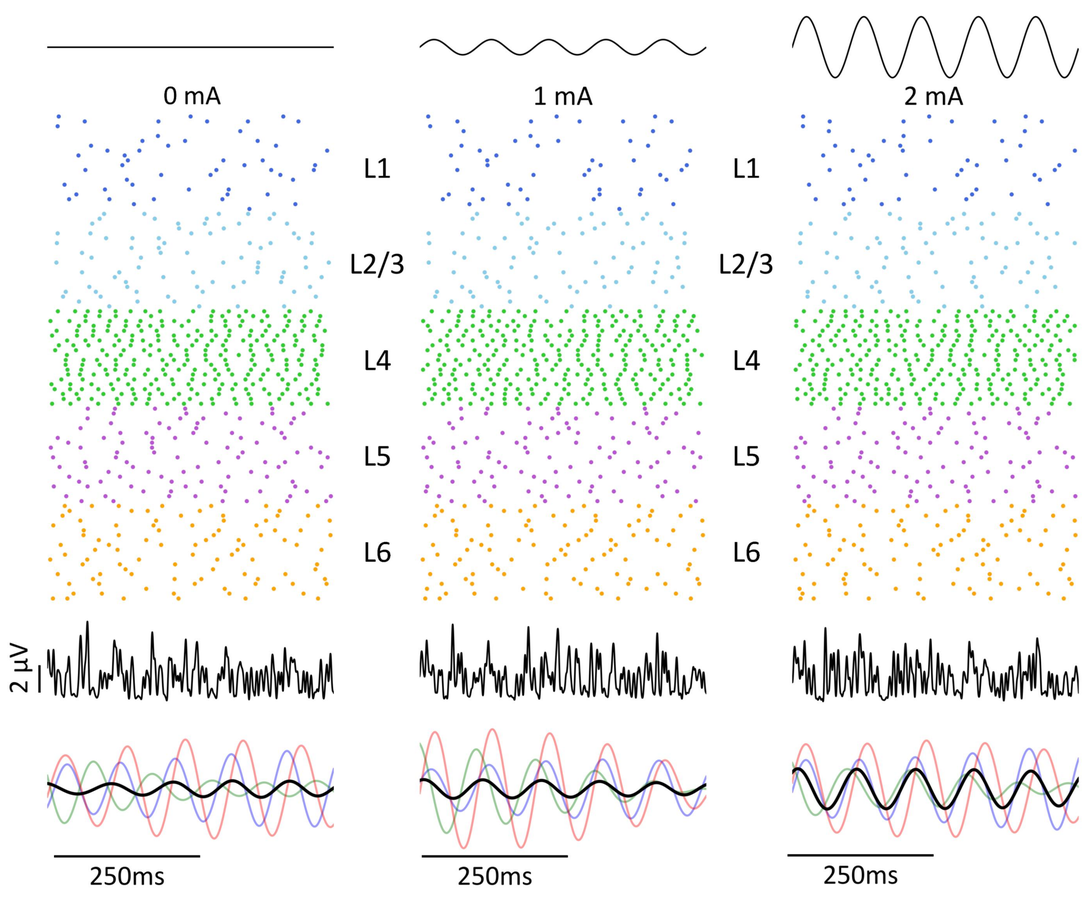
Materials and Methods

Components of the cortical microcircuit

- Implemented synaptic connections between cells with parameters based on [3]



Modeling cortical oscillations and tACS



Analysis of neural entrainment

▶ **Analysis of network-level entrainment**

- **Cross-correlation** of 20 LFP signals quantified synchronization across different tACS conditions (0, 1, 2 mA)

$$r_{xy}(0) = \frac{\sum_{i=1}^{N}(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^{N}(X_i - \bar{X})^2 \sum_{i=1}^{N}(Y_i - \bar{Y})^2}}$$

▶ **Analysis of single-neuron level entrainment**

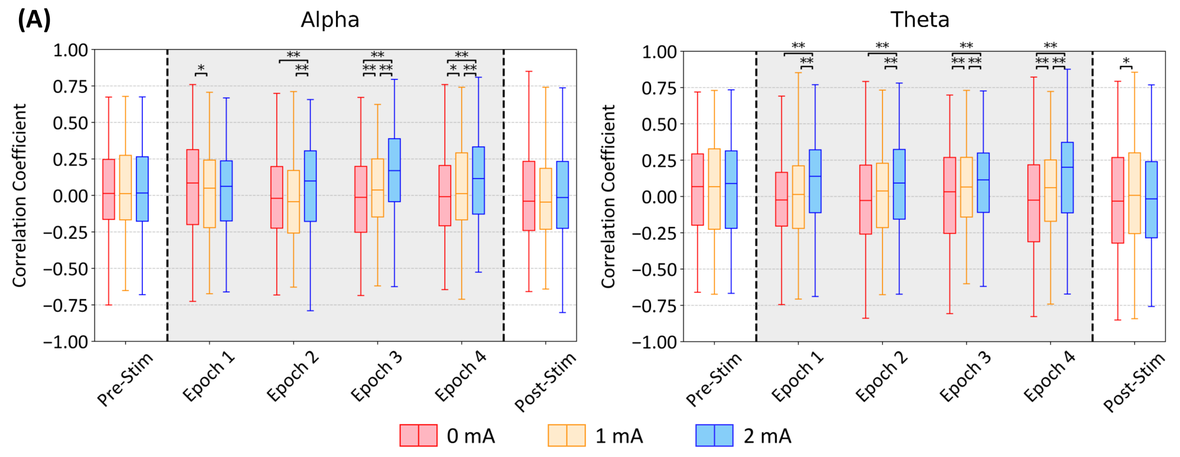
- **Phase locking value (PLV)** was used to quantify whether an individual neuron's spiking activity was synchronized with the tACS waveform

$$PLV = \left| \frac{\sum_{k=1}^{N} e^{i\theta_k}}{N} \right|$$

Results

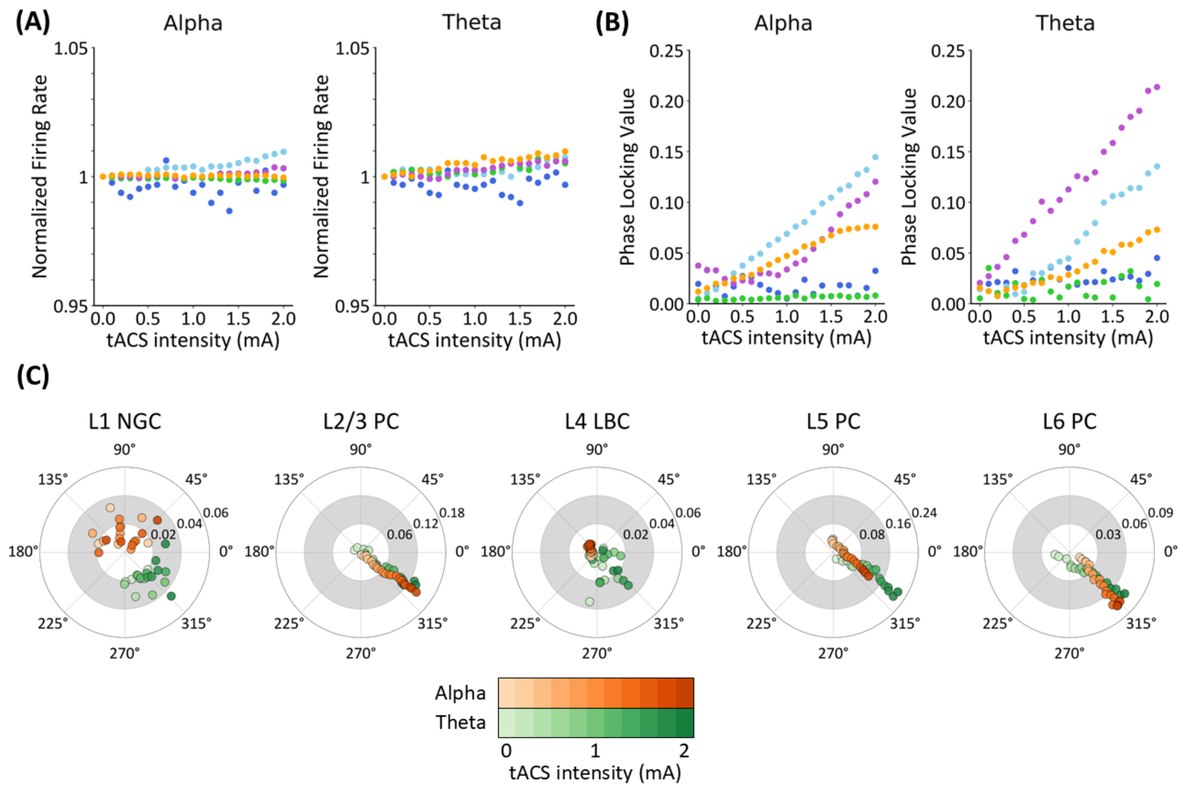
Pairwise LFP cross-correlation analysis

- Box plots display correlation coefficients across different stimulation intensities



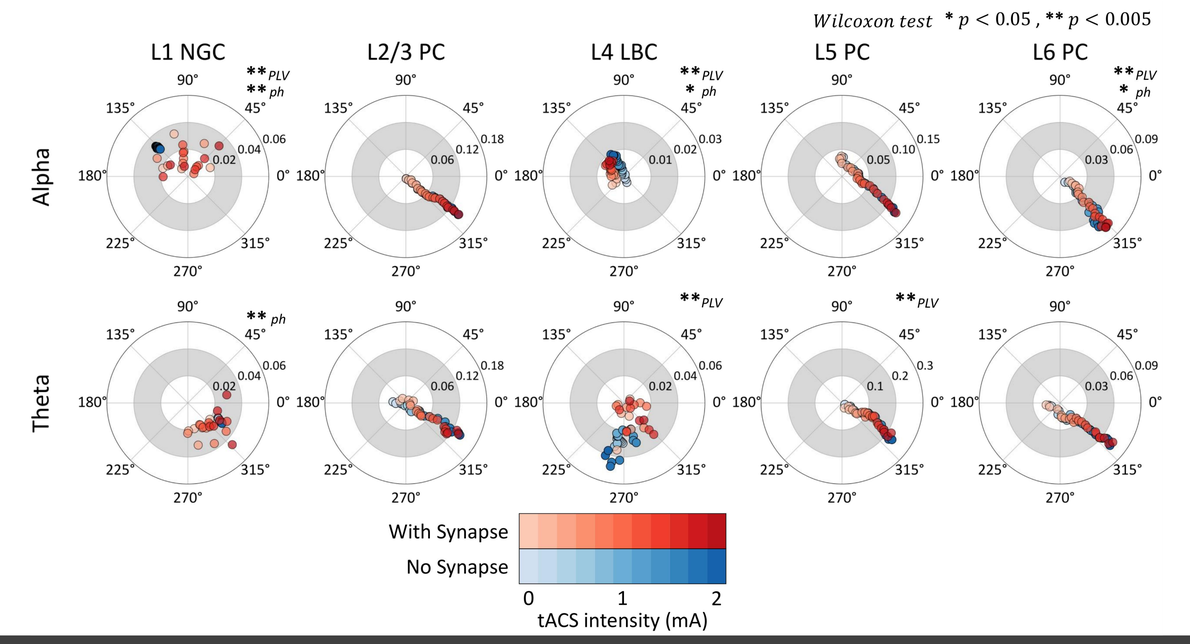
Neural response to tACS at the cellular level

- Pyramidal cells demonstrated a stronger phase-locking effect than the interneurons



Influence of network dynamics on neural entrainment

- Polar plots display the PLV and preferred phase across tACS intensities between a synaptically connected (red) and a disconnected (blue) microcircuit



Discussion

- tACS can either weaken the synchronization or more strongly entrain the neural response regarding the endogenous oscillations and the tACS intensities
- tACS response is suggested to result from the interplay between neuron morphology and network dynamics
- This work may reinforce the importance of neuronal morphology in shaping responses to tACS, as highlighted in recent research
- This work may support utilizing tACS to develop personalized therapies considering individual variability and the brain's endogenous neural oscillations

References

1. Chung et al., 2022 "Key factors in the cortical response to transcranial electrical stimulations-A multi-scale modeling study"
2. Cakan and Obermayer, 2020 "Biophysically grounded mean-field models of neural populations under electrical stimulation"
3. Markram et al., 2015 "Reconstruction and Simulation of Neocortical Microcircuitry"

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